



# SDLC Diagrams - Mermaid Codes & Report Locations



## Overview

This document contains all Mermaid codes for SDLC diagrams to be inserted in the IEEE report.



## DIAGRAM LOCATIONS IN REPORT

| Diagram             | Figure No.    | Chapter   | Section             | Page (Approx) |
|---------------------|---------------|-----------|---------------------|---------------|
| System Architecture | Fig 3.1       | Chapter 3 | 3.3 System Design   | ~Page 9-10    |
| Use Case Diagram    | Fig 3.2       | Chapter 3 | 3.3 System Design   | ~Page 10      |
| ER Diagram          | Fig 3.3       | Chapter 3 | 3.4 Database Design | ~Page 11-12   |
| Component Diagram   | Fig 3.5 (NEW) | Chapter 3 | 3.3 System Design   | ~Page 10      |
| Sequence Diagram    | Fig 3.6 (NEW) | Chapter 3 | 3.3 System Design   | ~Page 11      |
| Activity Diagram    | Fig 4.3 (NEW) | Chapter 4 | 4.2 Core Features   | ~Page 16      |



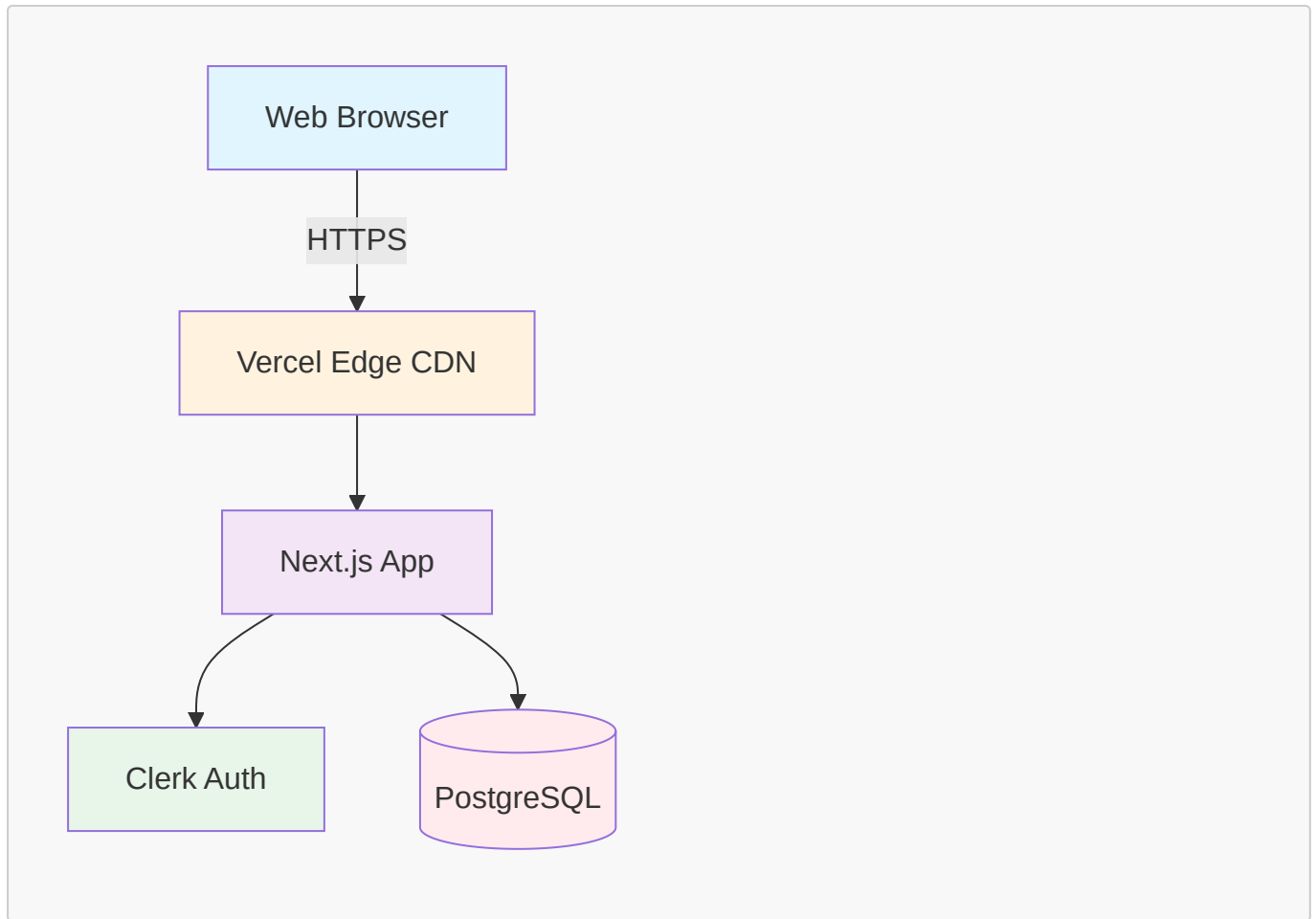
## HOW TO USE IN DRAW.IO

1. Open **Draw.io** (<https://app.diagrams.net/>)
2. Click **Arrange** → **Insert** → **Advanced** → **Mermaid**
3. **Copy** the Mermaid code from below
4. **Paste** into the Mermaid dialog
5. Click **Insert** to generate diagram
6. **Export** as PNG (File → Export as → PNG)
7. **Insert** the PNG image in report at the placeholder location



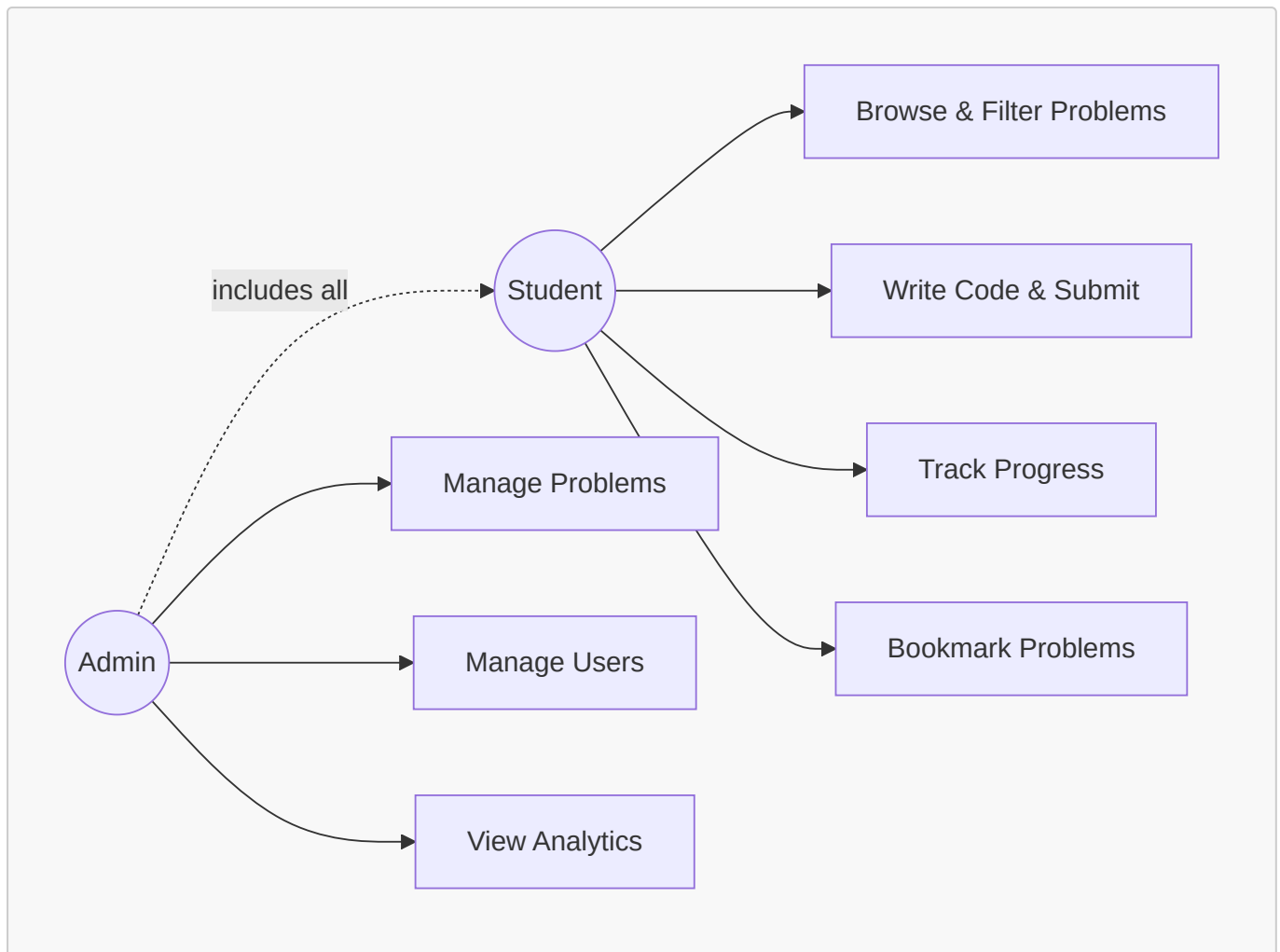
## MERMAID CODE #1: System Architecture Diagram (Fig 3.1)

**Location:** Chapter 3.3 - System Design



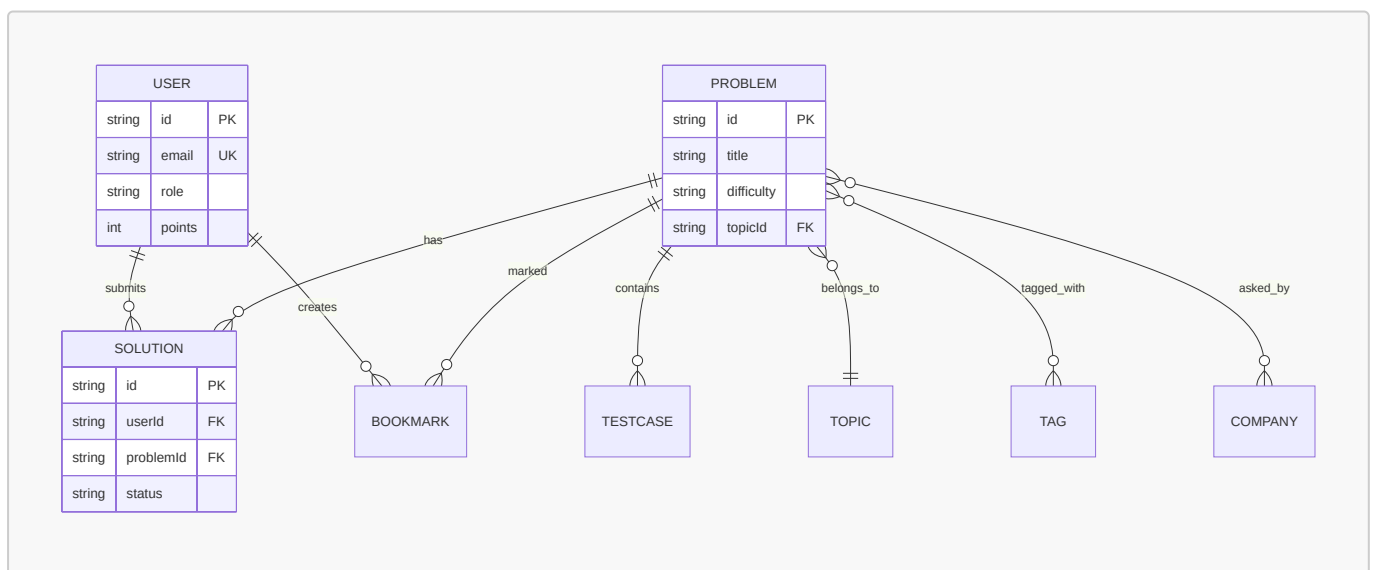
## 🎨 MERMAID CODE #2: Use Case Diagram (Fig 3.2)

**Location:** Chapter 3.3 - System Design



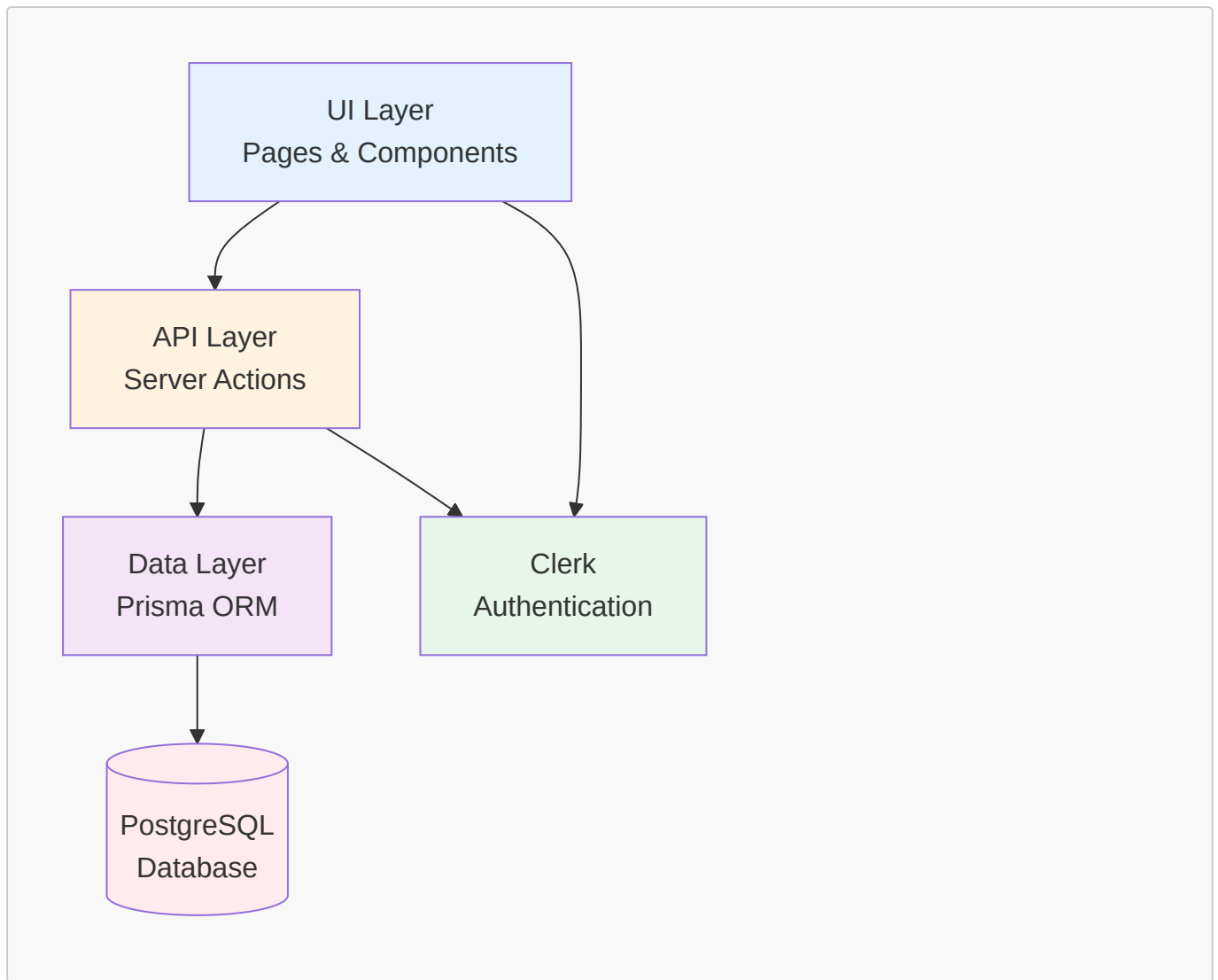
### 🎨 MERMAID CODE #3: ER Diagram (Fig 3.3)

**Location:** Chapter 3.4 - Database Design



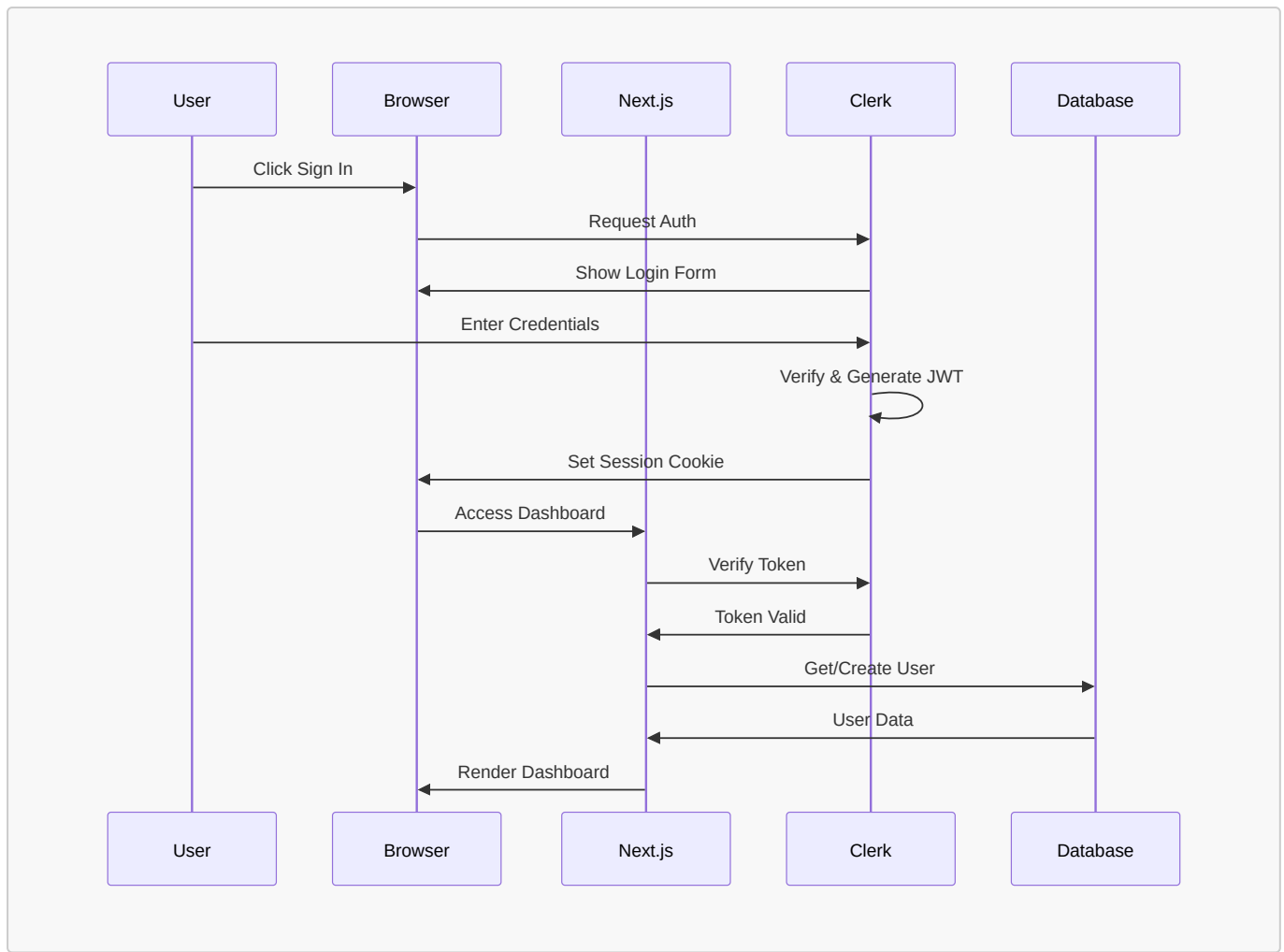
### 🎨 MERMAID CODE #4: Component Diagram (Fig 3.5)

**Location:** Chapter 3.3 - System Design (OPTIONAL - add if space available)



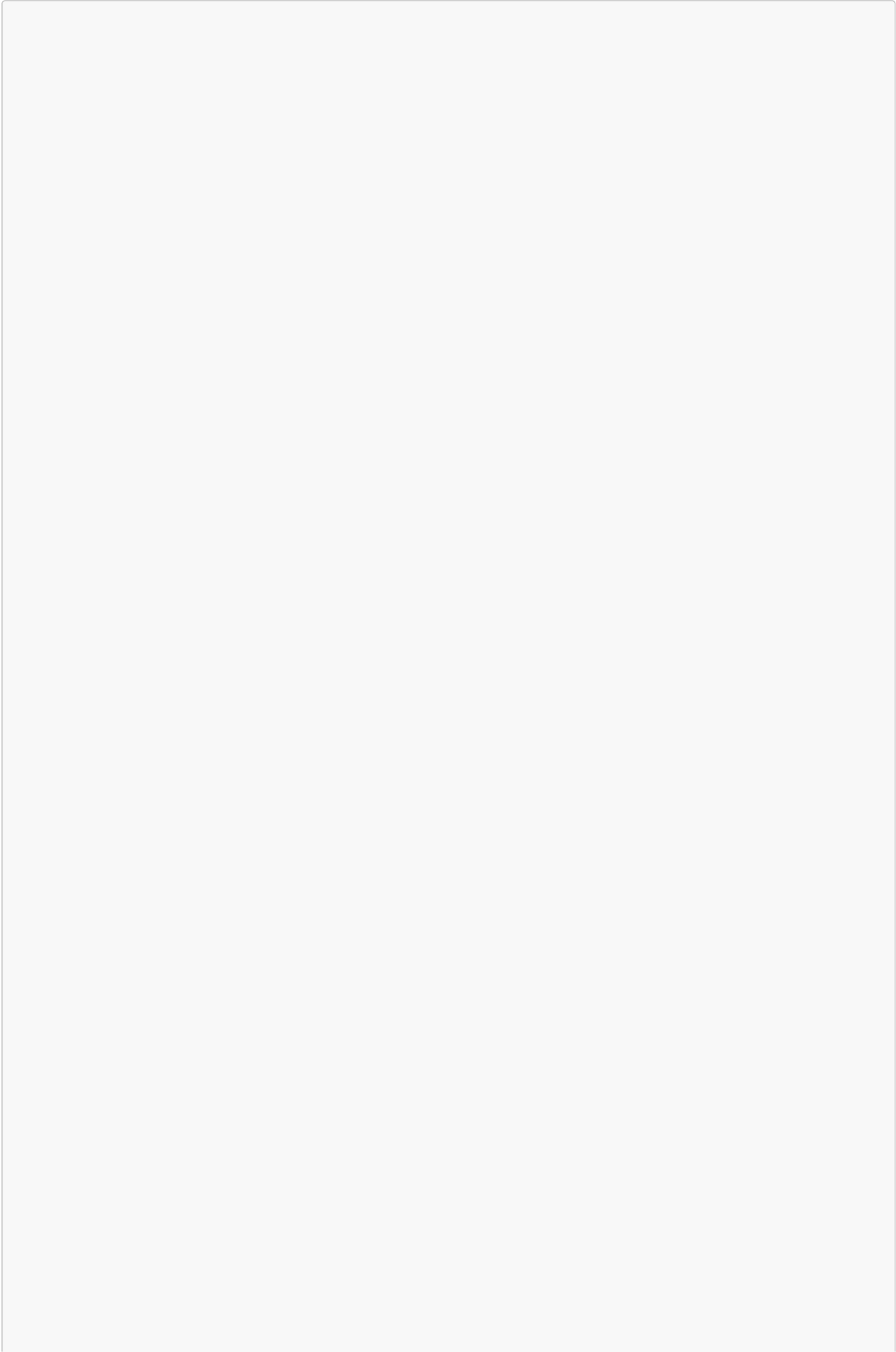
## MERMAID CODE #5: Sequence Diagram - Authentication (Fig 3.6)

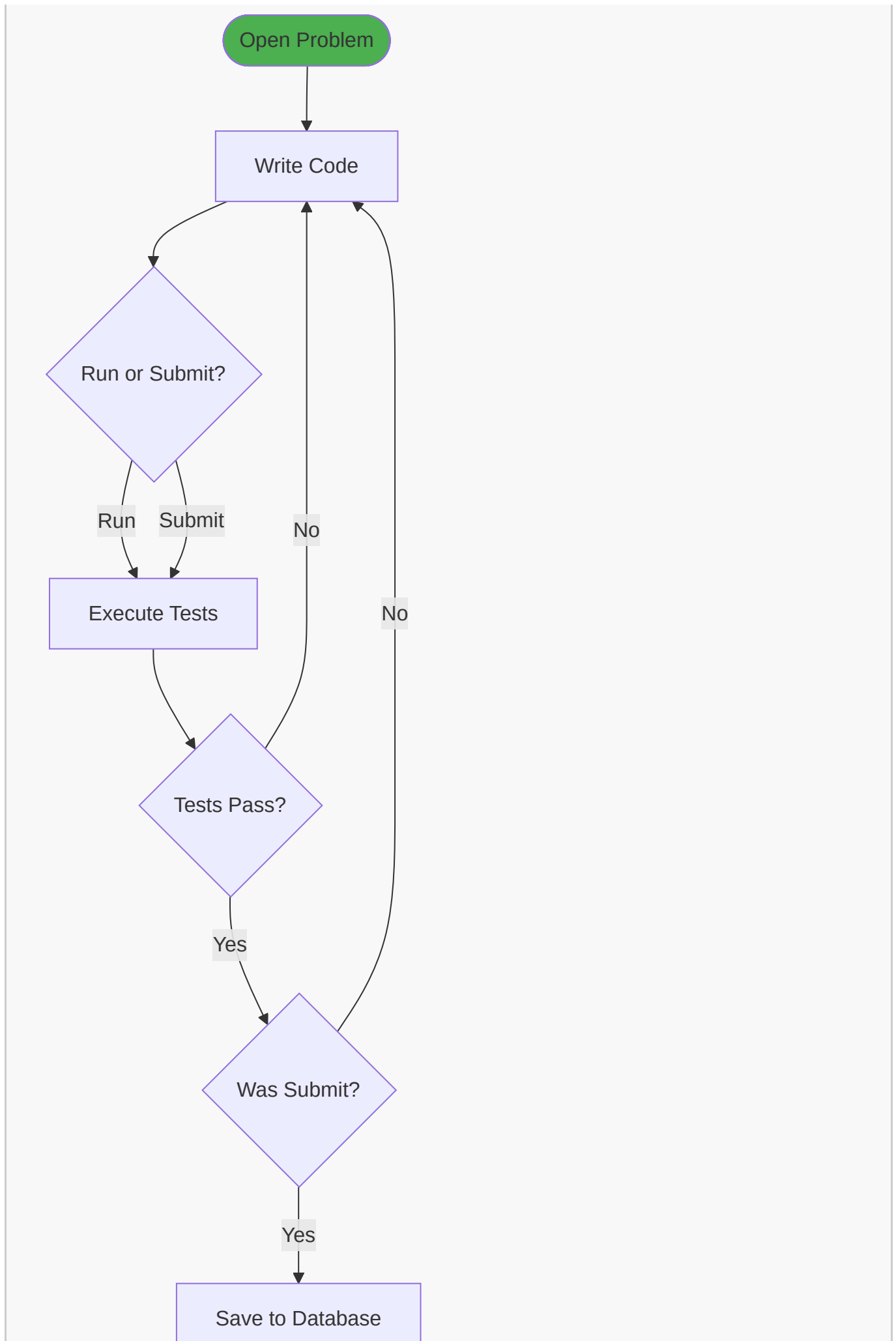
**Location:** Chapter 3.3 - System Design (OPTIONAL - add if space available)

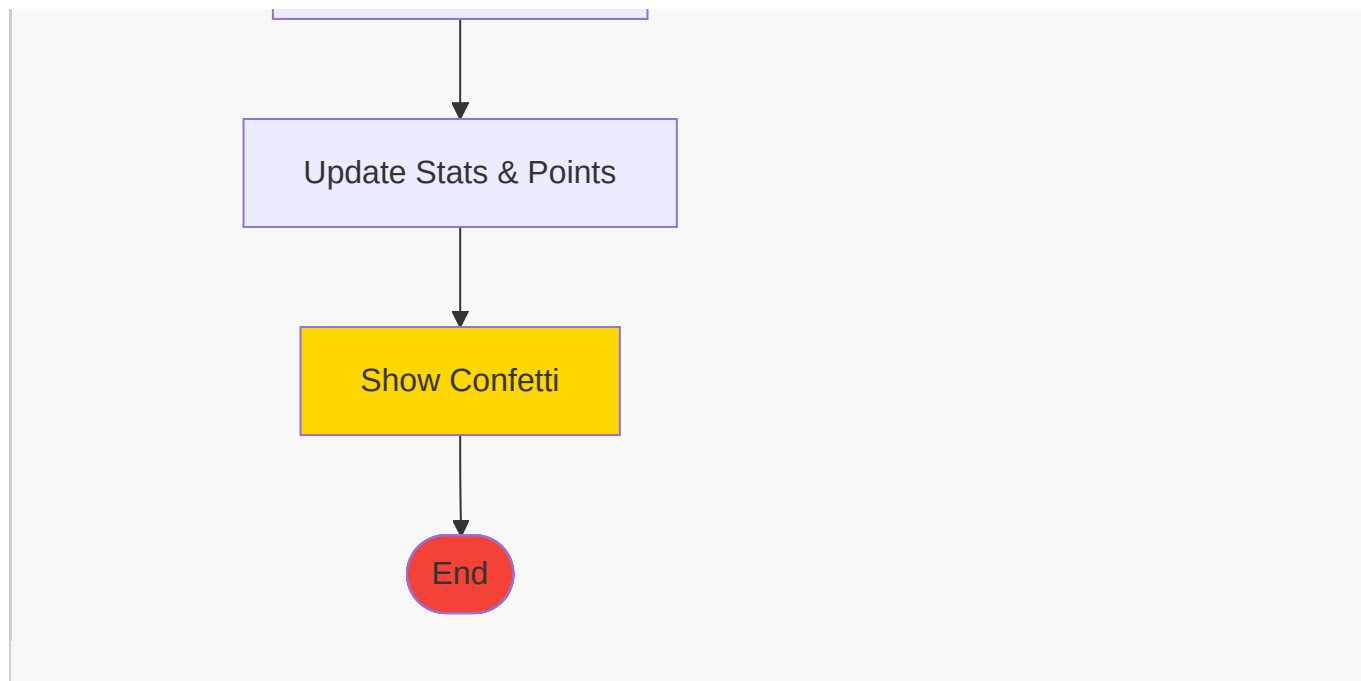


## 🎨 MERMAID CODE #6: Activity Diagram - Code Submission (Fig 4.3)

**Location:** Chapter 4.2 - Core Features Implementation (OPTIONAL - add if space available)







## ✓ MUST-HAVE DIAGRAMS (Top Priority)

These 3 diagrams are ESSENTIAL for SDLC compliance:

1. **Figure 3.1 - System Architecture** ✓ MUST ADD
2. **Figure 3.2 - Use Case Diagram** ✓ MUST ADD
3. **Figure 3.3 - ER Diagram** ✓ MUST ADD

## 📌 OPTIONAL DIAGRAMS (If Space Available)

Add these if report has extra space:

4. **Figure 3.5 - Component Diagram** (Good to have)
5. **Figure 3.6 - Sequence Diagram** (Good to have)
6. **Figure 4.3 - Activity Diagram** (Good to have)

## 🎯 PLACEHOLDER LOCATIONS IN REPORT

Search for these texts in the report to find where to insert diagrams:

1. "[INSERT DIAGRAM HERE: Use Mermaid Code #1 - System Architecture]"
2. "[INSERT DIAGRAM HERE: Use Mermaid Code #2 - Use Case Diagram]"
3. "[INSERT DIAGRAM HERE: Use Mermaid Code #3 - ER Diagram]"

## 📏 DIAGRAM SIZE RECOMMENDATIONS

When exporting from Draw.io:

- **Width:** 600-700 pixels (fits well in report)
- **Height:** Auto (maintain aspect ratio)



- **Format:** PNG with transparent background
  - **DPI:** 300 (high quality for printing)
- 

## CHECKLIST

- ☐ Copy Mermaid Code #1 (System Architecture)
  - ☐ Paste in Draw.io → Insert → Mermaid
  - ☐ Export as PNG (600px width)
  - ☐ Insert in report at Figure 3.1 placeholder
  - ☐ Repeat for Use Case Diagram (Fig 3.2)
  - ☐ Repeat for ER Diagram (Fig 3.3)
  - ☐ Verify all diagrams are visible in report
  - ☐ Update LIST OF FIGURES in Table of Contents
- 

## DONE!

After adding all 3 must-have diagrams, your report will have complete SDLC documentation! 